

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

UCI Forecasting Feature Importance & Interpretability Report

1. Feature Importance Summary

The top predictive features driving next-hour load forecasts:

- load_t (Importance: 0.7578)
- cos_hour (Importance: 0.0770)
- rolling_max_24 (Importance: 0.0432)
- sin_hour (Importance: 0.0401)
- rolling_mean_24 (Importance: 0.0328)
- hour (Importance: 0.0183)
- rolling_std_24 (Importance: 0.0073)
- rolling_min_24 (Importance: 0.0052)
- rolling_std_168 (Importance: 0.0030)
- load_t_minus_12 (Importance: 0.0022)

2. Key Domain Interpretations

- Lag 0 (load_t) and Lag 1 (load_t_minus_1)**: The most immediate prior load values dominate predictions, reflecting strong autoregressive continuity.
- Daily Seasonality (load_t_minus_24)**: 24-hour lag features strongly anchor forecasts to same-time-yesterday usage.
- Rolling Averages (rolling_mean_24, rolling_mean_168)**: Provide smooth baseline trend indicators for multi-day shifts.